

ES-FA / BTP Code Notes

Source file notes after latest smoke run

May 5, 2026

1 Training Core

File: `p1_training/training_core.py`.

Role: shared training loop for the baseline and experiments.

Why it matters: all software comparisons pass through the same training path, so the current 95.70% accuracy and 56.48% sparsity numbers are comparable across baseline and experiment runs.

Key interfaces: `load_config()` and `train_from_config()`.

2 SNN Model

File: `p1_training/model.py`.

Role: defines the proposal-aligned SNN model.

Model used: MNIST input flattened to 784, hidden layers of 128 and 64 LIF neurons, and 10 output classes.

Notes: the model supports temporal spike encoding and fake quantization for hardware-oriented experiments.

3 Hardware Proxy Estimators

File: `p2_hardware_model/estimators.py`.

Role: provides lightweight hardware proxy estimates.

Current use: the latest run uses these estimators to compare baseline energy proxy (22,604,295.76) with the best hardware-aware run (4,592,863.17).

Key interfaces:

- `estimate_spike_cost()`
- `estimate_memory_access()`
- `estimate_energy_proxy()`
- `estimate_latency_proxy()`

4 Experiments

Folder: `experiments/`.

Role: holds experiment configs and runners.

Latest smoke run: ran `exp1_hardware_aware_loss` and `exp5_dataflow_adaptation`. The best score came from `exp1_hardware_aware_loss`.

5 Hardware And Validation

Folders: `hardware/` and `hardware_validation/`.

Role: Verilog RTL and local validation wrappers.

Current status: RTL and validation scripts are present. Fresh hardware execution was skipped in the latest smoke run, but the flow is ready for xsim and Vivado when those stages are enabled.

6 System And Deployment Layer

Folders: `system_layer/` and `deployment/`.

Role: simple runtime demo layer.

Current demo: the CLI can answer control queries, run a small math example, and route a signal classification request through the adaptive policy.

7 Analysis

Folder: `analysis/`.

Role: creates ranking tables, plots, and evidence comparisons.

Latest output: 3 software rows, 4 existing hardware rows in the evidence schema, and 16 comparison rows in `results/analysis/evidence/`.

8 Output Artifacts

Folder: `output/`.

Role: presentation and report artifacts.

Latest generated files:

- `method_section.md`
- `experiment_section.md`
- `tables.md`
- `key_findings.md`
- `paper_draft.md`